

Reproducibility

Reproducibility in GNN is not an aspiration layered on top of the system; it is the operating contract that the pipeline enforces. The 0–24 processing steps are deterministic given a model specification and a target directory, and every published claim in this manuscript is traceable to a command that regenerates the underlying artifact. This section lists only commands that exist in the repository, so that a reader with a clean checkout can reproduce the pipeline, the validation gates, and this manuscript itself.

Pipeline Smoke Run I

The fastest way to confirm a working installation is to drive the full pipeline over the discrete model family without invoking the optional LLM steps:

```
uv run python src/gnn/main.py --target-dir input/gnn_files/
```

This parses the discrete GNN files, runs visualization and rendering across the maintained backends, and writes all artifacts under the chosen output directory. The `--skip-llm` flag keeps the run hermetic and free of external API calls: the non-LLM steps all execute, the steps that would read the skipped LLM outputs record that as a warning, and the run exits 2 — the pipeline's documented warning code (0 success, 1 error, 2 warning) — rather than 0. To exercise every

Pipeline Smoke Run I

registered family rather than a single one, drive the manifest through the model-family acceptance gate given below: pointing `--target-dir` at `input/gnn_files` covers that tree's 11 corpus directories. All 9 registered family target directories lie inside that tree, so a single invocation reaches every registered family.

The discrete family exercises the categorical kind end to end. The continuous linear-Gaussian kind smoke-runs the same way, and the contrast between the two runs is itself a check of the per-kind contract:

Pipeline Smoke Run I

```
uv run python src/gnn/main.py --target-dir input/gnn_files/
```

This parses the continuous specifications, and the render and execute steps fan them out to the continuous-capable backends — where they render as linear-Gaussian models and run as filtering (and, for the closed-loop exemplar, belief-steering) programs — while the categorical-only backends record explicit unsupported statuses rather than failures. A reader comparing the two receipts sees the kind taxonomy behaving as described in sec. 3: same pipeline, same steps, per-kind rendering and execution reach.

GNN's reproducibility guarantees rest on a small set of strict, deterministic gates that bind the manuscript's quantitative claims to recomputable ledgers. The model-family acceptance gate runs the maintained families declared in the manifest and fails on any regression:

```
uv run python scripts/run_model_family_acceptance.py \
  --manifest input/model_family_manifest.json \
  --output-dir output/model_family_acceptance --strict
```

The semantic-fidelity gate verifies that a `parse` $\rightarrow$ `serialize` $\rightarrow$ `parse` round trip preserves variables, edges, dimensions, parameter shapes, equations, time semantics, and ontology mappings across the 9 model families; the cross-framework gate profiles the 7 maintained backends (PyMDP, RxInfer.jl, JAX, NumPyro, PyTorch, ActiveInference.jl, DisCoPy) — refusing any framework outside that set — and records explicit compatible and unsupported statuses rather than silently degrading. Both write their ledgers to an output directory of your choosing:

Validation Gates I

```
uv run python scripts/run_semantic_fidelity_gate.py \  
  --manifest input/model_family_manifest.json \  
  --output-dir output/semantic_fidelity --strict  
uv run python scripts/run_cross_framework_reliability.py \  
  --manifest input/model_family_manifest.json \  
  --output-dir output/cross_framework --strict
```

Validation Gates I

Under `--strict`, each gate exits non-zero on the first mismatch, so these commands double as assertions in an automated reproduction run. Code-quality reproducibility is enforced separately through the developer command reference: `just lint` runs the Ruff linter over `src` and `scripts`, and the broader `just quality` recipe chains formatting, terminology, documentation, type, and security checks for a full pre-commit gate.

This manuscript is itself a reproducible artifact. Every quantitative value in the prose — the pipeline step count, the family and backend counts, the source and test inventories — is a token rather than a hard-coded literal, and the deterministic producer regenerates all of them from the tracked files at the current commit:

```
uv run python scripts/z_generate_manuscript_variables.py
```

That command recomputes the `{{...}}` tokens, persists them to `output/data/manuscript_variables.json` for audit, and hydrates the manuscript sources into `output/manuscript/`. The manuscript's own figures are rebuilt from the same token map:

```
uv run python -m scripts.manuscript_build_figures
```

The hydrated sources are then rendered to PDF by the docxology template's render stage. That stage lives in a separate checkout, with this repository symlinked into it at `projects/active/GeneralizedNotationNotation`; run from the template root:

```
uv run --frozen python scripts/pipeline/stage_03_render.py  
      --project GeneralizedNotationNotation
```

The render needs a LaTeX installation providing the packages listed in `manuscript/preamble.md` plus `seqsplit`; the template guards `seqsplit` with `\IfFileExists`, so a missing copy degrades rather than failing the build.

Because the variables file is regenerated before rendering, the counts in the rendered PDF track the repository state at the commit recorded in `output/data/manuscript_variables.json` (f5ef5cf23): a code change that alters, for example, the test inventory (458 test files, 4905 test functions) propagates into the prose on the next regeneration without any manual editing.

Reproducibility Contract I

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- ▶ Do not cite results that cannot be regenerated or directly traced to a command in this repository.
- ▶ Keep generated outputs under `output/` and maintained manuscript source under `manuscript/`; treat everything in `output/` as disposable and regeneratable.
- ▶ Express every quantitative claim in the prose as a double-brace `{{...}}` token substituted by `scripts/z_generate_manuscript_variables.py`, never as a hard-coded number.
- ▶ Keep private data, credentials, and unpublished sensitive details out of the manuscript and out of version control.
- ▶ Record the exact verification commands — the smoke run, the acceptance and fidelity gates, and just `lint` — before marking this manuscript publication-ready.